Sampling and Per-Precision Histograms with nixnan

An Experiment on `rd_nixnan.cu`

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The Example, at a Glance

`rd_nixnan.cu`: a 1D reaction-diffusion PDE $\partial u/\partial t = D\,\partial^2 u/\partial x^2 + \lambda u$, explicit FTCS, 101-point grid, 1600 steps — the *same* kernel implemented once each in FP16, BF16, FP32, FP64.

How precision-sensitive is this example?

- The reaction term grows every value by $\sim$1.05$\times$ per step, *regardless of format* — so the only thing that differs between formats is how much exponent range each one has before it overflows.
- FP16's 5-bit exponent runs out at step 300; BF16/FP32/FP64's wider exponents are all still finite at step 1600 — same algorithm, same input, one format fails and three don't.

NixNan Feature Reference: BIN_SPEC_FILE

nixnan's own banner gives this example:

```
{ "count": 128, "bf16": [], "f16": [[0,5],[-4,-1]],
  "f32": [], "f64": [] }
```

- Each format key: a list of *unbiased signed exponent* ranges to watch. Here only `f16` watches anything: $[0, 5]$ and $[-4, -1]$.
- `count`: report every `count`-th occurrence in a watched range — `count:128` = "every 128th hit."
- **Zero and infinity are structurally excluded from every range** — a value that jumps straight from finite to `inf` in one step is invisible to this mechanism, no matter how ranges are drawn. (Matters later: this is exactly what happens here.)

Concept 1: Sampling — The Mechanism

Every run here uses `nixnan.so` directly ("straight nixnan"), no wrapper:

```
LD_PRELOAD=<path>/nixnan.so [ENV_VARS...] ./rd_nixnan
```

`SAMPLING=N` instruments only every *N*th launch of a repeated kernel — a knob trading overhead for observation completeness.

Verified before trusting it: an older comment in this repo claims *"SAMPLING is a no-op."* We re-tested directly (`SAMPLING=1` vs. `SAMPLING=100000`): exception counts dropped to exactly zero at the high value. **Confirmed fixed** in the current build.

Concept 1: Every Call vs. Once in 8

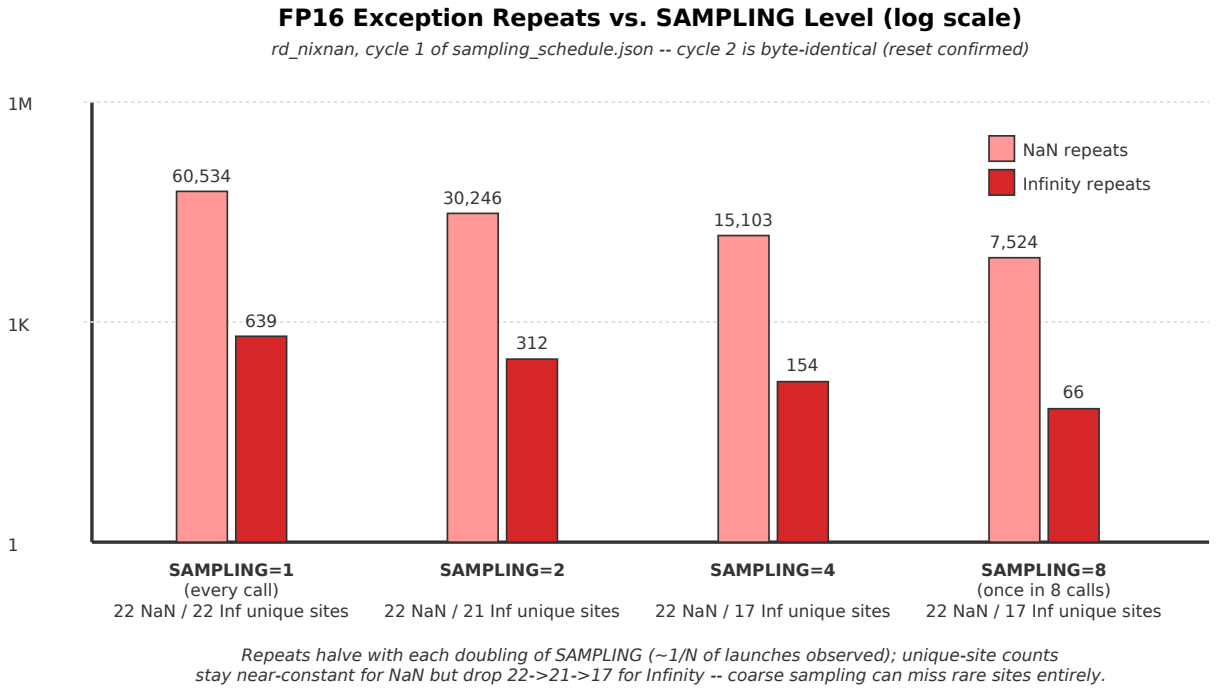
FP16 exception counts (the only format that overflows here):

	NaN (unique / repeats)	Infinity (unique / repe
`SAMPLING=1` (every call)	22 / 60,534	22 /
`SAMPLING=8`	22 / 7,524	17 /

Repeats drop $\sim 8\times$, as expected. But **5 distinct Infinity sites vanish entirely** ($22 \rightarrow 17$) — coarse sampling doesn't just under-count, it can miss whole bugs.

Concept 1: An Exponentially Decaying Schedule

`SAMPLING` is fixed for a whole process run, so a decaying rate is built as a sequence of short runs (`sampling_schedule.json`): step 4 times at spacing 1 $\rightarrow$ double to 2 $\rightarrow$ 4 $\rightarrow$ 8 $\rightarrow$ **reset**.



Every snap at a level is byte-identical; **cycle 2 exactly reproduces cycle 1** — the reset is provably clean. The TSV = 32 independent runs (4 levels $\times$ 4 snaps $\times$ 2 cycles), each just `SAMPLING=$level LD_PRELOAD=nixnan.so ./rd_nixnan` — no other setting changed from the Section 4 baseline.

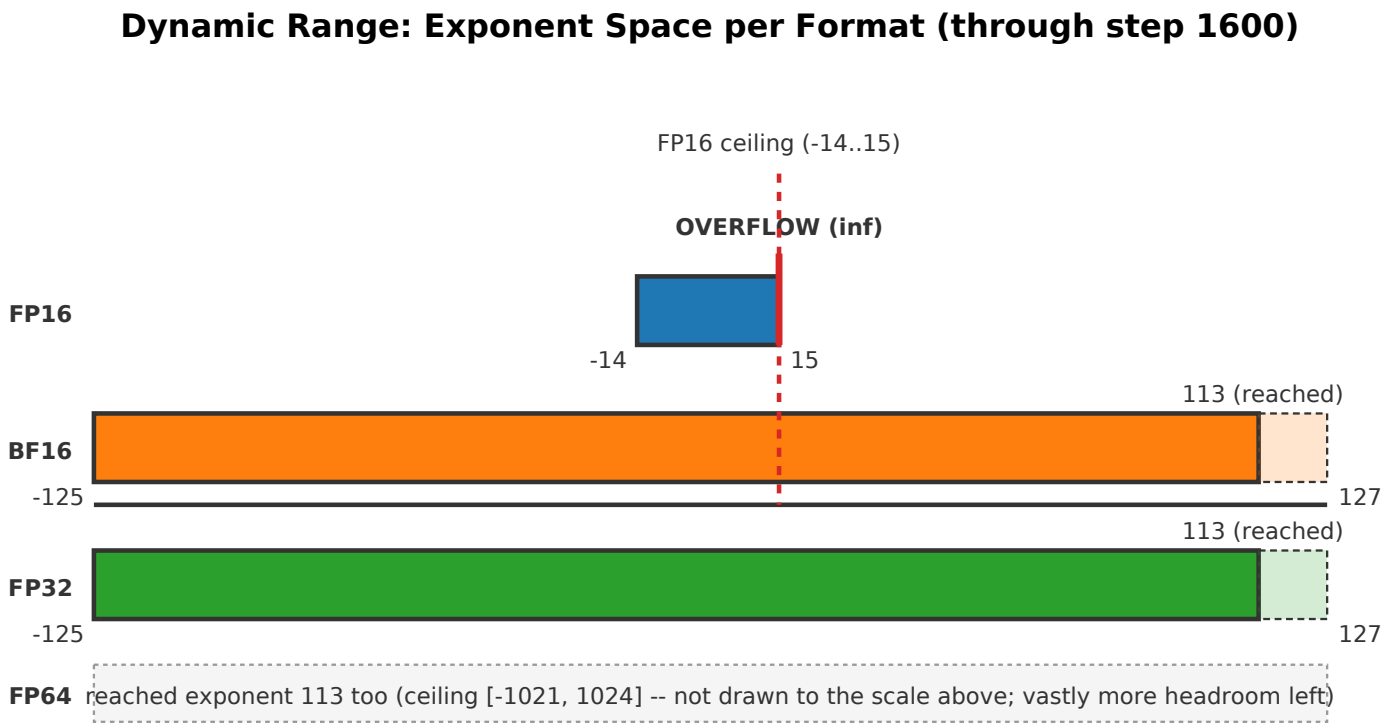
Concept 2: Histograms per Precision — The Mechanism

`HISTOGRAM=1` tracks the exponent of every instrumented FP value, independent of exception checking:

```
HISTOGRAM=1 LD_PRELOAD=nixnan.so ./rd_nixnan
```

At exit, nixnan reports the [min, max] exponent *actually observed* for each format — one number per precision bucket, no per-value log needed.

Concept 2: Results — Which Format Actually Failed



`f16` alone reports `inf`. `bf16`/`f32`/`f64` all reach the *identical* exponent, 113 — same trajectory, different ceilings, only one ceiling reached.

Concept 3: Making BF16 and FP32 Fail Too

Every result so far: FP16 fails, the other three don't. Can we make BF16/FP32 fail as well?

Tried: domain enlargement (bigger N). Does **not** work — the reaction term `lam * u[i]` grows every point by the same factor regardless of N/dx. Already confirmed earlier this session: `rd_nixnan_large.cu` (10× grid, N=1001) still only failed in FP16 at M=1600.

Works: more time steps. M is a compile-time macro:

```
nvcc -arch=sm_86 -DM=2000 rd_nixnan.cu -o rd_nixnan_extended
LD_PRELOAD=nixnan.so ./rd_nixnan_extended
```

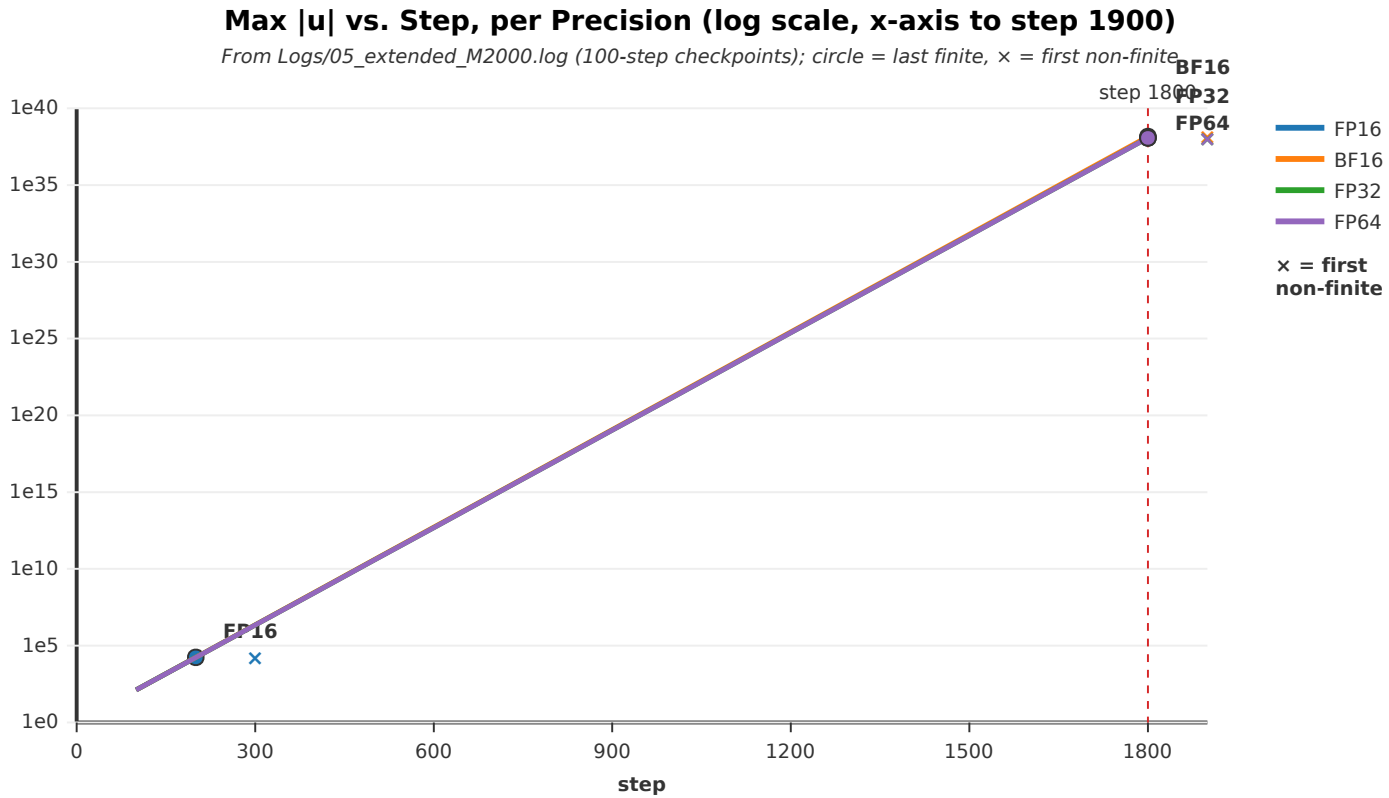
Concept 3: Results at M=2000

Format	First non-finite	NaN u/r	Inf u/r
FP16	step 300	22 / 78134	22 / 639
BF16	step 1900	15 / 8975	6 / 76
FP32	step 1900	19 / 10966	8 / 90
FP64	step 1900	see caveat	

BF16 and FP32 now both show real nixnan-classified exceptions.

Honest caveat: FP64 also goes non-finite (program's own check, step 1900), but nixnan's classifier reports *zero* for FP64 — even with more margin (M=2500). A real, unresolved gap, reported as observed.

Concept 3: Where Infinity Actually Comes From



FP16 (blue) departs the shared curve at step 300. BF16/FP32/FP64 stay numerically **indistinguishable** through step 1800 (1.1675×10^{38}, all three, to 4 sig. figs.) and are all non-finite by step 1900.

Concept 3: The Exact CUDA Statement

Rebuilt with `-g -lineinfo` so nixnan's `LINE_INFO=1` could resolve real source locations (not just SASS):

```
error [infinity] detected ... instruction FADD R0, R0, R7
  ; in function rd_step_fp32 at rd_nixnan.cu:96 of type f32
```

Line 96: `float lap = (u[i+1] + u[i-1]) - 2.f * u[i];` — **Infinity first appears in the diffusion term, not the reaction term.** (FP16's `__h*` intrinsics never resolve a line in this build — a real nixnan limitation; same conclusion by structural analogy.)

Inf → NaN, confirmed to the instruction:

```
FADD R0,R0,R7 ; u[i+1]+u[i-1] -> operand already Infinity
FADD R7,R6,R6 ; 2.f*u[i] -> operand already Infinity
FADD R7,R0,-R7 ; (sum) - 2.f*u[i] -> RESULT is NaN (inf - inf)
```

Matches the source file's own long-standing comment (`inf - 2*inf + inf -> NaN`) — now tool-verified, not hand-derived.

What nixnan Made Visible Here (1/2)

1. **Sampling trades overhead for completeness**, not just verbosity — coarse sampling can drop entire exception sites, not only reduce their reported repeat count.
2. **A schedule can be built around a static env var** via orchestration, and — because the program is deterministic — that schedule's reset is independently verifiable.
3. **Per-format exponent histograms turn "FP16 probably fails" into a specific, quantified claim**: FP16 alone overflows; the other three formats are proven to share one common growth trajectory up to a shared checkpoint exponent.

What nixnan Made Visible Here (2/2)

4. **nixnan can adjudicate a hypothesis about why a format fails first**: domain enlargement is ruled out by the kernel's own growth term; running longer is confirmed, with nixnan itself newly reporting real BF16/FP32 exceptions — and surfacing an FP64 classification gap.
5. **Line-level attribution turns a hand-derived guess into a tool-verified fact**: with `-lineinfo`, nixnan pinpoints the exact statement (the Laplacian) and instruction where $\infty - \infty$ produces `NaN` — confirming a mechanism the source comments could only predict by hand.